

Integrality of the sequence OEIS A276175

Abstract

We prove that the order-four rational recurrence

$$a_{n+4} a_n = (a_{n+1} + 1)(a_{n+2} + 1)(a_{n+3} + 1), \quad a_0 = a_1 = a_2 = a_3 = 1,$$

produces only integers. A rational number is an integer iff it has nonnegative valuation at every prime, so the statement splits into two independent parts. For odd primes the valuations are controlled by an elementary “atom” induction. The remaining obstruction, at the prime 2, is settled by a 44-step 2-adic return lemma over the tube $1 + 16\mathbb{Z}_2$, whose only computational core is a finite, decidable check in the rings $\mathbb{Z}/2^m\mathbb{Z}$.

1 Statement and strategy

Let $(a_n)_{n \geq 0}$ be defined by

$$a_0 = a_1 = a_2 = a_3 = 1, \quad a_{n+4} a_n = (a_{n+1} + 1)(a_{n+2} + 1)(a_{n+3} + 1) \quad (n \geq 0). \quad (1)$$

A straightforward induction shows every a_n is a well-defined positive rational (each $a_n > 0$, so no division by zero occurs).

Theorem 1. $a_n \in \mathbb{Z}$ for every $n \geq 0$.

A nonzero rational r is an integer if and only if $v_p(r) \geq 0$ for every prime p , where v_p is the p -adic valuation. Since each a_n is a positive rational, Theorem 1 is equivalent to

$$v_p(a_n) \geq 0 \quad \text{for every prime } p \text{ and every } n \geq 0. \quad (2)$$

We prove (2) in two parts, treated by completely different methods:

- Odd primes p (Section 2): an elementary induction on auxiliary integers $t_p(n)$ attached to the shifted sequence $b_n = a_n + 1$. This yields $a_n \in \mathbb{Z}[\frac{1}{2}]$.
- The prime 2 (Section 3): a 2-adic return lemma showing $v_2(a_n)$ equals an explicit nonnegative 44-periodic word.

Together they give (2) and hence Theorem 1 (Section 4).

Throughout, write $b_n = a_n + 1$. From (1), $a_{n+4}a_n = b_{n+1}b_{n+2}b_{n+3}$, and expanding $b_{n+4} = a_{n+4} + 1$ gives the identity used repeatedly below:

$$b_{n+4} a_n = a_n + b_{n+1}b_{n+2}b_{n+3}. \quad (3)$$

All a_n, b_n are positive, hence nonzero, so every valuation of a product or quotient below splits additively.

2 Odd primes: an atom induction

Fix an odd prime p and abbreviate $v_p = v$. Define integers $t_n := t_p(n)$ by

$$t_0 = t_1 = t_2 = t_3 = 0, \quad t_{n+4} = v(b_{n+4}) - t_n, \quad (4)$$

so that, by construction,

$$v(b_{n+4}) = t_{n+4} + t_n. \quad (5)$$

Thus t_n is the “atom” of p -adic valuation freshly deposited at index n ; (5) says the valuation of b_{n+4} is the sum of the atom it carries and the atom four steps back.

Proposition 1. *For every $n \geq 0$ the following hold:*

- (I) $t_n \geq 0$;
- (II) $v(a_{n+4}) = t_{n+1} + t_{n+2} + t_{n+3}$;
- (III) if $t_{n+4} > 0$ then $t_{n+1} = t_{n+2} = t_{n+3} = 0$.

Property (III) is a *burst-confinement*: valuation can appear only in isolated bursts, never adjacent to a previous one. We first record the consequence.

$v(a_n) \geq 0$ for all n . As p ranges over all odd primes, $a_n \in \mathbb{Z}[\frac{1}{2}]$.

Proof. For $n \leq 3$, $a_n = 1$ and $v(a_n) = 0$. For $n \geq 4$, (II) and (I) give $v(a_n) = t_{n-3} + t_{n-2} + t_{n-1} \geq 0$. Hence no odd prime divides the denominator of a_n . $\square$

Proof of Proposition 1. We argue by strong induction on n , proving (I)–(III) simultaneously. The base indices $n = 0, \dots, 7$ use the explicit values

$$\begin{aligned} a_4 = 2^3, \quad a_5 = 2^2 \cdot 3^2, \quad a_6 = 2 \cdot 3^2 \cdot 37, \quad a_7 = 3^2 \cdot 37 \cdot 667, \\ b_4 = 3^2, \quad b_5 = 37, \quad b_6 = 23 \cdot 29, \quad b_7 = 2^5 \cdot 11 \cdot 631, \end{aligned}$$

from which (I)–(III) are checked directly (for an odd prime p , each t_j with $4 \leq j \leq 7$ equals v of the corresponding odd integer, hence ≥ 0 , and a positive t_j pins p to one of the listed odd factors, forcing the three preceding atoms to vanish). Assume now $n \geq 4$ and that (I)–(III) hold for all smaller indices.

(II). Taking v of $a_{n+4}a_n = b_{n+1}b_{n+2}b_{n+3}$ and using (5) termwise,

$$v(a_{n+4}) + v(a_n) = (t_{n+1} + t_{n-3}) + (t_{n+2} + t_{n-2}) + (t_{n+3} + t_{n-1}).$$

By the inductive form of (II), $v(a_n) = t_{n-3} + t_{n-2} + t_{n-1}$, and subtracting gives $v(a_{n+4}) = t_{n+1} + t_{n+2} + t_{n+3}$.

(III). Suppose $t_{n+4} > 0$. By (5) and (I), $v(b_{n+4}) = t_{n+4} + t_n \geq t_{n+4} > 0$, so $p \mid b_{n+4}$, i.e. $a_{n+4} \equiv -1 \pmod{p}$. Since $a_{n+4} = b_{n+4} - 1$ with $v(b_{n+4}) > 0 = v(1)$, the ultrametric inequality is an equality and $v(a_{n+4}) = 0$. By (II), $t_{n+1} + t_{n+2} + t_{n+3} = 0$, and each summand is ≥ 0 by (I), so all three vanish.

(I). We must show $v(b_{n+4}) \geq t_n$, i.e. $t_{n+4} \geq 0$. If $t_n = 0$ this is immediate: $b_{n+4} = 1 + a_{n+4}$ with $v(a_{n+4}) \geq 0$ (by (II) and (I) at smaller indices), so $v(b_{n+4}) \geq \min(0, v(a_{n+4})) = 0$.

Now suppose $T := t_n > 0$. Applying (III) at index $n - 4$ gives $t_{n-3} = t_{n-2} = t_{n-1} = 0$, whence by (II)

$$v(a_n) = t_{n-3} + t_{n-2} + t_{n-1} = 0,$$

so a_n is a p -adic unit. We bound the four summands of the identity (3), rewritten (using $b_j = a_j + 1$) as

$$b_{n+4}a_n = \underbrace{b_n}_{(1)} + \underbrace{a_{n+3}b_{n+2}b_{n+1}}_{(2)} + \underbrace{a_{n+2}b_{n+1}}_{(3)} + \underbrace{a_{n+1}}_{(4)}. \quad (6)$$

Each has valuation $\geq T$:

- (1) $v(b_n) = t_n + t_{n-4} = T + t_{n-4} \geq T$ by (5) and (I);
- (4) $v(a_{n+1}) = t_{n-2} + t_{n-1} + t_n = T$ by (II) and the vanishing above;
- (3) $v(a_{n+2}) = t_{n-1} + t_n + t_{n+1} \geq T$ and $v(b_{n+1}) \geq 0$;
- (2) $v(a_{n+3}) = t_n + t_{n+1} + t_{n+2} \geq T$ and $v(b_{n+2}), v(b_{n+1}) \geq 0$.

(For (3),(4) note $v(a_{n+1}), v(a_{n+2}) \geq 0$ and the displayed sums are $\geq T$ because $t_n = T$ and the other atoms are ≥ 0 ; the values used for $v(a_{n+1}), v(a_{n+2}), v(a_{n+3})$ come from (II).) By the ultrametric inequality the right side of (6) has valuation $\geq T$, so $v(b_{n+4} a_n) \geq T$. As $v(a_n) = 0$, we conclude $v(b_{n+4}) \geq T = t_n$, i.e. $t_{n+4} \geq 0$. This closes the induction. $\square$

3 The prime 2: a 44-step return

The behaviour at $p = 2$ is genuinely 2-adic and does not reduce to a congruence in a fixed modulus: naive term-by-term 2-adic estimates lose precision at every step. The remedy is to control an entire 2-adic neighbourhood of the orbit simultaneously and to exhibit an explicit period.

Let

$$w = (w_0, \dots, w_{43}), \quad t = (t_0, \dots, t_{43})$$

be the 44-periodic words

$$\begin{aligned} w &= (0, 0, 0, 0, 3, 2, 1, 0, 2, 3, 4, 0, 0, 0, 0, 3, 2, 1, 0, 2, 3, 4, \\ &\quad 0, 0, 0, 0, 4, 3, 2, 0, 1, 2, 3, 0, 0, 0, 0, 4, 3, 2, 0, 1, 2, 3), \\ t &= (1, 1, 1, 1, 0, 0, 0, 5, 0, 0, 0, 2, 1, 1, 1, 0, 0, 0, 5, 0, 0, 0, \\ &\quad 2, 1, 1, 2, 0, 0, 0, 5, 0, 0, 0, 1, 1, 1, 2, 0, 0, 0, 5, 0, 0, 0), \end{aligned}$$

extended 44-periodically to all indices. A direct check gives the *balance law*

$$w_{n+4} + w_n = t_{n+1} + t_{n+2} + t_{n+3} \quad (n \geq 0). \quad (7)$$

Lemma 1 (44-step return lemma). *Let $A_0, A_1, A_2, A_3 \in 1 + 16\mathbb{Z}_2$ and extend (A_n) by*

$$A_{n+4}A_n = (1 + A_{n+1})(1 + A_{n+2})(1 + A_{n+3}).$$

Then $A_0, \dots, A_{47}$ are all defined and units of $\mathbb{Q}_2^\times$, with

$$v_2(A_n) = w_n \quad (0 \leq n \leq 43),$$

and moreover

$$A_{44}, A_{45}, A_{46}, A_{47} \in 1 + 16\mathbb{Z}_2.$$

The point of the lemma is that the hypothesis and the last conclusion have the *same* form: a window of four elements of $1 + 16\mathbb{Z}_2$ is mapped, after 44 steps, to another such window. Thus the lemma may be iterated.

Proof. The statement is uniform in the tube $1 + 16\mathbb{Z}_2$, so we prove it once for the generic point. Work in the power-series ring $R = \mathbb{Z}_2[[z_0, z_1, z_2, z_3]]$ and set

$$A_i = 1 + 16z_i \quad (0 \leq i \leq 3),$$

so that specialising $z_0, \dots, z_3 \in \mathbb{Z}_2$ recovers an arbitrary admissible window. We claim each A_n factors as

$$A_n = 2^{w_n} U_n, \quad U_n \in R^\times, \quad (8)$$

with the units satisfying, by (7),

$$U_{n+4} = U_n^{-1} H_{n+1} H_{n+2} H_{n+3}, \quad H_n := \frac{1 + 2^{w_n} U_n}{2^{t_n}}. \quad (9)$$

Indeed, substituting (8) into the recurrence and dividing by $2^{w_{n+4}+w_n} = 2^{t_{n+1}+t_{n+2}+t_{n+3}}$ yields (9), provided each H_n is a well-defined unit and each U_n is invertible. So the lemma reduces to the finite assertion that, for $0 \leq n \leq 47$,

- (a) the division by 2^{t_n} defining H_n is exact and $H_n \in R^\times$;
- (b) $U_n \in R^\times$ (so U_n^{-1} exists);
- (c) at every phase with $w_n = 0$ one has $v_2(1 + U_n(z)) = t_n$ for all $z \in \mathbb{Z}_2^4$;
- (d) $U_{44}, U_{45}, U_{46}, U_{47} \equiv 1 \pmod{16R}$.

Here $U_n(z)$ denotes the specialisation and $U_n(0)$ its constant term.

This is a finite, coefficientwise computation over the finite rings $\mathbb{Z}/2^m\mathbb{Z}$. One tracks each U_n as a polynomial in $z_0, \dots, z_3$ with coefficients known modulo 2^{p_n} (a running precision), starting from $U_i = 1 + 16z_i$ at precision $p_i = 34$. The four ring operations needed for (9)—multiplication, addition, division by a power of 2 (exact, hence precision-preserving up to the shift), and inversion of a 1 mod 2 series by Newton iteration—are all exact modulo 2^p . Writing $U_n = \sum_\alpha u_{n,\alpha} z^\alpha$ and $\mu_n = \min_{\alpha \neq 0} v_2(u_{n,\alpha})$, condition (c) is verified by the two facts

$$v_2(1 + U_n(0)) = t_n \quad \text{and} \quad \mu_n \geq t_n + 1 \quad (10)$$

at each phase with $w_n = 0$: the first fixes the constant term's valuation, the second forces every nonconstant coefficient of $1 + U_n$ to lie deeper, so that $v_2(1 + U_n(z)) = t_n$ for all $z \in \mathbb{Z}_2^4$. At a phase with $w_n > 0$ the element $1 + 2^{w_n} U_n$ is a unit and $t_n = 0$. Finally (d) is the statement $U_n(0) \equiv 1 \pmod{16}$ with $\mu_n \geq 4$ for $n = 44, \dots, 47$.

The verification is elementary but too large to reproduce by hand; carrying it out (by exact integer arithmetic, and independently machine-checked in a proof assistant) confirms (a)–(d). For orientation, the decisive phases—those with $w_n = 0$ —give the following constant terms and depths, from which (10) is read off:

n	0	1	2	3	7	11	12	13	14	18	22	23
t_n	1	1	1	1	5	2	1	1	1	5	2	1
$U_n(0) \bmod 64$	1	1	1	1	31	43	9	21	9	31	11	45
μ_n	4	4	4	4	6	3	4	4	3	7	4	4

n	24	25	29	33	34	35	36	40	44	45	46	47
t_n	1	2	5	1	1	1	2	5	1	1	1	1
$U_n(0) \bmod 64$	61	43	31	17	21	41	3	31	17	49	17	49
μ_n	4	4	7	3	4	4	3	6	4	4	4	4

Starting from precision 34, the running precisions at $U_{44}, \dots, U_{47}$ are 9, 8, 7, 6, comfortably above the 4 bits the return congruence (d) requires. This proves the lemma. $\square$

$$v_2(a_n) = w_n \geq 0 \text{ for all } n \geq 0.$$

Proof. The initial window $a_0 = a_1 = a_2 = a_3 = 1$ lies in $1 + 16\mathbb{Z}_2$, and (a_n) obeys the recurrence of Lemma 1 (indeed (1) is exactly it, with $A_n = a_n$). The lemma gives $v_2(a_n) = w_n$ for $0 \leq n \leq 43$ and places the window $(a_{44}, a_{45}, a_{46}, a_{47})$ back in $1 + 16\mathbb{Z}_2$. Applying the lemma again from index 44, then 88, and so on, yields $v_2(a_n) = w_{n \bmod 44}$ for every n . As all $w_n \geq 0$, the claim follows. $\square$

4 Conclusion

By Corollary 2, $a_n \in \mathbb{Z}[\frac{1}{2}]$: no odd prime divides its denominator. By Corollary 3, $v_2(a_n) \geq 0$: no power of 2 divides its denominator either. Hence a_n has denominator 1, i.e. $a_n \in \mathbb{Z}$, proving Theorem 1. $\square$

Remark (The cluster embedding). The auxiliary sequence in Section 2 can be replaced by a structural observation. Define (x_n) by

$$x_0 = 4, \ x_1 = \frac{1}{2}, \ x_2 = 4, \ x_3 = \frac{1}{2}, \quad x_{n+4}x_n = 1 + x_{n+1}x_{n+2}x_{n+3}.$$

Then $x_{n+4}x_n = 1 + x_{n+1}x_{n+2}x_{n+3}$ shows (a'_n) with $a'_n := x_{n+1}x_{n+2}x_{n+3}$ satisfies exactly (1) with the same initial values, so $a_n = x_{n+1}x_{n+2}x_{n+3}$. This recurrence is a period-one cluster recurrence (mutation at one vertex of the totally-ordered 4-vertex quiver, which is recovered after a cyclic relabelling); the Laurent phenomenon then gives $x_n \in \mathbb{Z}[x_0^{\pm 1}, \dots, x_3^{\pm 1}] = \mathbb{Z}[\frac{1}{2}]$, hence $a_n \in \mathbb{Z}[\frac{1}{2}]$. This reproves Corollary 2 and explains the provenance of the problem, but is not needed for the argument above.

Remark. The valuation sequence $v_2(a_n)$ is exactly 44-periodic (Corollary 3); no shorter period occurs (for instance $v_2(a_{15}) = 3$ while $v_2(a_{26}) = 4$). This is why a return lemma over a full period, rather than a single-step congruence, is required at the prime 2.